

Continuous Integration and Installation of Tomcat

Continuous Integration and Installation of Tomcat with MavenAnsibleWebApp

MavenAnsibleWebApp is a DevOps project that demonstrates Continuous Integration (CI) and automated deployment of a web application using Apache Maven, Ansible, Jenkins, and Apache Tomcat.

About Tomcat

Apache Tomcat is an open-source web server and servlet container developed by the Apache Software Foundation. It is mainly used to run Java-based web applications such as JSP (Java Server Pages) and Servlets.

Tomcat acts as an application server where Java web applications are deployed and executed. It receives client requests through a web browser, processes the application logic, and sends responses back to the user.

Features of Tomcat

- Lightweight and fast web server
- Supports Java Servlets and JSP
- Easy deployment of WAR files
- Open-source and platform-independent
- Widely used for web application hosting

Role of Tomcat in DevOps

In DevOps, Tomcat is commonly used for:

- Hosting Java web applications
- Automated deployment through CI/CD pipelines
- Testing and staging environments
- Continuous delivery workflows

Working of MavenAnsibleWebApp

1. Developer writes and updates application code.
2. Code is pushed to a Git repository.
3. Jenkins automatically detects changes and starts the CI process.
4. Maven compiles the code, runs tests, and creates a WAR file.
5. Ansible installs and configures Tomcat automatically on the server.
6. The WAR file is deployed into the Tomcat server.
7. Tomcat hosts and runs the web application for users.

Advantages

- Automated build and deployment
- Faster software delivery
- Reduced manual errors
- Easy server configuration
- Efficient Continuous Integration process

Applications

- Java web application hosting
- Automated deployment pipelines
- DevOps laboratory projects
- Cloud-based web application management

This project demonstrates how Maven, Jenkins, Ansible, and Tomcat work together in a DevOps environment to automate application development, server setup, deployment, and continuous integration processes efficiently.

Steps to execute in a project:

Step 1:

Open terminal

- `sudo -i` → Switches the user to root (administrator) mode with full system privileges.
- `sudo apt update && sudo apt upgrade -y` → Updates the package list and upgrades all installed packages automatically without asking for confirmation.

```
sudo: upgrade: command not found
root@vaishnavi-virtualbox:~# sudo apt update && sudo apt upgrade -y
Hit:1 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:4 https://ppa.launchpadcontent.net/mozillateam/ppa/ubuntu jammy InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:6 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
173 packages can be upgraded. Run 'apt list --upgradable' to see them.
E: dpkg was interrupted, you must manually run 'sudo dpkg --configure -a' to correct the problem.
```

Step 2:

Download Apache Tomcat

- `cd /tmp` → Changes the current directory to the system's temporary folder.
- `wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat9.0.95.tar.gz` → Downloads Apache Tomcat 9.0.95 compressed file from the official Apache archive to the system.

```
root@vaishnavi-virtualbox:~# cd /tmp
root@vaishnavi-virtualbox:/tmp# wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
--2026-04-13 19:55:50-- https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.95/bin/apache-tomcat-9.0.95.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12715996 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.95.tar.gz'

apache-tomcat-9.0.95.tar.gz      100%[=====] 12.13M  454KB/s  in 20s
2026-04-13 19:56:11 (634 KB/s) - 'apache-tomcat-9.0.95.tar.gz' saved [12715996/12715996]
```

Step 3:

Create Directory for Tomcat Installation

- `cd ..` → Moves one level up from the current directory.
- `cd /opt` → Navigates to the /opt directory used for installing optional software.
- `mkdir tomcat` → Creates a new directory named tomcat to store the Tomcat files.
- `ls` → Lists directory contents to verify that the tomcat folder is created successfully.

```

root@vaishnavi-virtualbox:/tmp# cd ..
root@vaishnavi-virtualbox:/# cd opt
root@vaishnavi-virtualbox:/opt# ls
google tomcat
root@vaishnavi-virtualbox:/opt# cd tomcat
root@vaishnavi-virtualbox:/opt/tomcat# cd ..
root@vaishnavi-virtualbox:/opt# cd tmp
-bash: cd: tmp: No such file or directory
root@vaishnavi-virtualbox:/opt#

```

Step 4: Extract Apache Tomcat Files

- `sudo tar -xvzf apache-tomcat-9.0.95.tar.gz -C /opt/tomcat --strip-components=1` →

Extracts the downloaded Tomcat archive into /opt/tomcat directory by removing the toplevel folder and placing all files directly inside it.

```

apache-tomcat-9.0.95/webapps/host-manager/css/manager.css
apache-tomcat-9.0.95/webapps/host-manager/images/asf-logo.svg
apache-tomcat-9.0.95/webapps/host-manager/images/tomcat.svg
apache-tomcat-9.0.95/webapps/host-manager/index.jsp
apache-tomcat-9.0.95/webapps/manager/META-INF/context.xml
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/401.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/403.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/404.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/connectorCerts.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/connectorCiphers.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/connectorTrustedCerts.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/sessionDetail.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/jsp/sessionsList.jsp
apache-tomcat-9.0.95/webapps/manager/WEB-INF/web.xml
apache-tomcat-9.0.95/webapps/manager/css/manager.css
apache-tomcat-9.0.95/webapps/manager/images/asf-logo.svg
apache-tomcat-9.0.95/webapps/manager/images/tomcat.svg
apache-tomcat-9.0.95/webapps/manager/index.jsp
apache-tomcat-9.0.95/webapps/manager/status.xsd
apache-tomcat-9.0.95/webapps/manager/xform.xsl
apache-tomcat-9.0.95/bin/catalina.sh
apache-tomcat-9.0.95/bin/ciphers.sh
apache-tomcat-9.0.95/bin/configtest.sh
apache-tomcat-9.0.95/bin/daemon.sh
apache-tomcat-9.0.95/bin/digest.sh
apache-tomcat-9.0.95/bin/makebase.sh
apache-tomcat-9.0.95/bin/setclasspath.sh
apache-tomcat-9.0.95/bin/shutdown.sh
apache-tomcat-9.0.95/bin/startup.sh
apache-tomcat-9.0.95/bin/tool-wrapper.sh
apache-tomcat-9.0.95/bin/version.sh

```

Step 5:

Configure Tomcat Server

- `cd /opt/tomcat/conf` → Navigates to the Tomcat configuration directory.
- `ls` → Lists configuration files inside the directory.
- `gedit server.xml` → Opens the main Tomcat configuration file for editing.
- Port Change (8080 → 9090) → Modifies the default Tomcat port to 9090 to avoid conflicts with other services.

```

root@vaishnavi-virtualbox:/opt/tomcat# ls
bin BUILDING.txt conf CONTRIBUTING.md lib LICENSE logs NOTICE README.md RELEASE-NOTES RUNNING.txt temp webapps work
root@vaishnavi-virtualbox:/opt/tomcat# cd conf/
root@vaishnavi-virtualbox:/opt/tomcat/conf# ls
catalina.policy context.xml jaspic-providers.xsd server.xml tomcat-users.xsd
catalina.properties jaspic-providers.xml logging.properties tomcat-users.xml web.xml

```

```

52 so you may not define subcomponents such as "Valves" at this level,
53 Documentation at /docs/config/service.html
54 -->
55 <Service name="Catalina">
56
57 <!--The connectors can use a shared executor, you can define one or more named thread
58 pools-->
59 <!--
60 <Executor name="tomcatThreadPool" namePrefix="catalina-exec-"
61     maxThreads="150" minSpareThreads="4"/>
62 -->
63
64 <!-- A "Connector" represents an endpoint by which requests are received
65 and responses are returned. Documentation at :
66 Java HTTP Connector: /docs/config/http.html
67 Java AJP Connector: /docs/config/ajp.html
68 APR (HTTP/AJP) Connector: /docs/apr.html
69 Define a non-SSL/TLS HTTP/1.1 Connector on port 8080
70 -->
71 <Connector port="8080" protocol="HTTP/1.1"
72     connectionTimeout="20000"
73     redirectPort="8443"
74     maxParameterCount="1000"
75 />
76 <!-- A "Connector" using the shared thread pool-->
77 <!--
78 <Connector executor="tomcatThreadPool"
79     port="8080" protocol="HTTP/1.1"
80     connectionTimeout="20000"
81     redirectPort="8443"
82     maxParameterCount="1000"
83 />
84 -->
85 <!-- Define an SSL/TLS HTTP/1.1 Connector on port 8443
86 This connector uses the NIO implementation. The default
87 SSLImplementation will depend on the presence of the APR/native

```

Step 6:

Configure Tomcat User Access

- gedit tomcat-users.xml → Opens the Tomcat user configuration file.
- Add user with role → Creates a user (admin) with password and assigns manager-gui role to access the Tomcat Manager dashboard

```

See the License for the specific language governing permissions and
limitations under the License.
-->
<tomcat-users xmlns="http://tomcat.apache.org/xml"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://tomcat.apache.org/xml tomcat-users.xsd"
    version="1.0">
    <user username="admin" password="Admin" roles="manager-gui"/>
<!--
By default, no user is included in the "manager-gui" role required
to operate the "/manager/html" web application. If you wish to use this app,
you must define such a user - the username and password are arbitrary.

```

Step 7:

Start Tomcat Server

- cd /opt/tomcat/bin → Navigates to the Tomcat binary directory.
- ls → Lists all available startup and shutdown scripts.
- sh startup.sh → Starts the Apache Tomcat server successfully.


```

root@vaishnavi-virtualbox:/opt/tomcat/cont# cd ..
root@vaishnavi-virtualbox:/opt/tomcat# ls
bin BUILDING.txt conf CONTRIBUTING.md lib LICENSE logs NOTICE README.md RELEASE-NOTES RUNNING.txt temp webapps work
root@vaishnavi-virtualbox:/opt/tomcat# cd bin
root@vaishnavi-virtualbox:/opt/tomcat/bin# ls
bootstrap.jar ciphers.bat configtest.bat digest.sh setclasspath.sh startup.sh tool-wrapper.sh
catalina.bat ciphers.sh configtest.sh makebase.bat shutdown.bat tomcat-juli.jar version.bat
catalina.sh commons-daemon.jar daemon.sh makebase.sh shutdown.sh tomcat-native.tar.gz version.sh
catalina-tasks.xml commons-daemon-native.tar.gz digest.bat setclasspath.bat startup.bat tool-wrapper.bat
root@vaishnavi-virtualbox:/opt/tomcat/bin# sh startup.sh
Using CATALINA_BASE: /opt/tomcat
Using CATALINA_HOME: /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@vaishnavi-virtualbox:/opt/tomcat/bin# cd ..
root@vaishnavi-virtualbox:/opt/tomcat#

```

Step 8:

Verify Tomcat Installation

- Open Browser (<http://localhost:9090>) → Confirms Tomcat server is running successfully by displaying the default Tomcat page.
- Access Manager (<http://localhost:9090/manager/html>) → Logs into the Tomcat Manager dashboard to manage and deploy web applications.

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/9.0.95

If you're seeing this, you've successfully installed Tomcat. Congratulations!

Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

[Servlet Status](#)

[Manager App](#)

[Host Manager](#)

Developer Quick Start

- [Tomcat Sites](#)
- [Building & A&A](#)
- [Examples](#)
- [Servlet Specifications](#)
- [First Web Application](#)
- [JDBC Databases](#)
- [Tomcat Versions](#)

Managing Tomcat

For security, access to the `manager` subapp is restricted. Users are defined in:

```

CATALINA_HOME/conf/tomcat-users.xml

```

In Tomcat 9.0 access to the manager application is split between different users. [Read more...](#)

- [Release Notes](#)
- [Changelog](#)
- [Migration Guide](#)
- [Security Notices](#)

Documentation

- [Tomcat 9.0 Documentation](#)
- [Tomcat 9.0 Configuration](#)
- [Tomcat Wiki](#)

Find additional important configuration information in:

```

CATALINA_HOME/RUNNING.txt

```

Developers may be interested in:

- [Tomcat 9.0 Bug Database](#)
- [Tomcat 9.0 Javadocs](#)
- [Tomcat 9.0 Get Involved at Github](#)

Getting Help

FAQ and Mailing Lists

The following mailing lists are available:

- tomcat-users**
Important announcements, releases, security vulnerability notifications. (Low volume)
- tomcat-dev**
Developer support and discussion
- tomcat-catalina**
User support and discussion for Apache Tomcat
- tomcat-jsp**
Development mailing list, including general messages

Other Downloads

- [Tomcat Connectors](#)
- [Tomcat Native](#)

Other Documentation

- [Servlet Connector](#)
- [JSP & JSTL](#)

Get Involved

- [Overview](#)
- [Getting Started](#)

More Downloads

- [Tomcat](#)
- [JDK](#)

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Sign in

http://localhost:9090

Username

Password

Cancel
Sign in




Tomcat Web Application Manager

Manager

[List Applications](#)
[HTML Manager Help](#)
[Manager HTML](#)
[Server Status](#)

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	Yes	0	Start Stop Reload Undeploy Engine resources: with file 0 / 0 minutes
/servlet	None specified	Servlet Documentation	Yes	0	Start Stop Reload Undeploy Engine resources: with file 0 / 0 minutes
/examples	None specified	Servlet and JSP Examples	Yes	0	Start Stop Reload Undeploy Engine resources: with file 0 / 0 minutes
/host-manager	None specified	Tomcat Host Manager Application	Yes	0	Start Stop Reload Undeploy Engine resources: with file 0 / 0 minutes
/manager	None specified	Tomcat Manager Application	Yes	1	Start Stop Reload Undeploy Engine resources: with file 0 / 0 minutes

Deploy

Deploy directory on MM at the located on server:

Context Path:

Version (for parallel deployment):

XML Configuration file path:

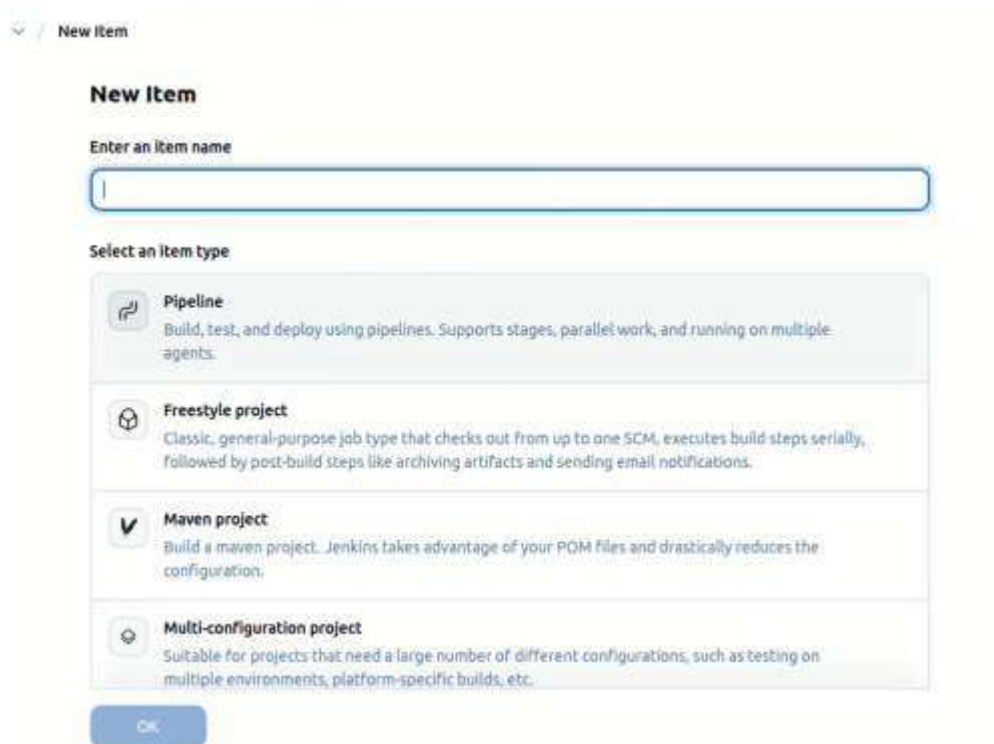
URL to Previous context:

Second Phase

Depicting Jenkins Pipeline

Follow the given steps to deploy the project on Jenkins.

create a Jenkins pipeline job that will use the Jenkinsfile from your project repository
go to the Jenkins home page and click on “New Item” to create a job, enter the item name select Pipeline as the project type.



The 'New Item' dialog box in Jenkins. It has a title 'New Item' and a subtitle 'Enter an item name'. Below the subtitle is a text input field. Underneath is a section 'Select an item type' with four options: 'Pipeline' (with a Jenkins logo icon), 'Freestyle project' (with a gear icon), 'Maven project' (with a checkmark icon), and 'Multi-configuration project' (with a gear icon). Each option has a brief description. At the bottom is an 'OK' button.

New Item

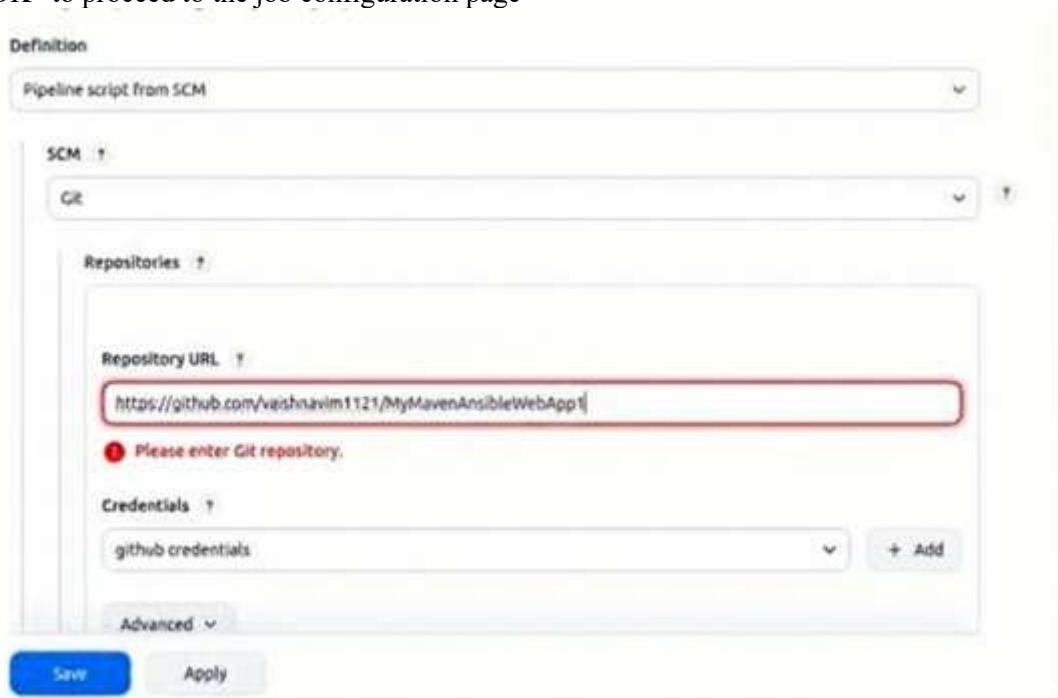
Enter an item name

Select an item type

- Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
- Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
- Maven project**
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
- Multi-configuration project**
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.

OK

This option allows you to build, test, and deploy applications using a Jenkinsfile. After selecting the pipeline type, click “OK” to proceed to the job configuration page



The Jenkins Pipeline Configuration page. It has a 'Definition' section with a dropdown menu set to 'Pipeline script from SCM'. Below this is an 'SCM' section with a dropdown menu set to 'Git'. Underneath is a 'Repositories' section with a 'Repository URL' input field containing 'https://github.com/vaishnavim1121/MyMavenAnsibleWebApp1'. Below the URL is a red error message: 'Please enter Git repository.' Underneath is a 'Credentials' section with a dropdown menu set to 'github credentials' and an 'Add' button. At the bottom are 'Save' and 'Apply' buttons.

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

https://github.com/vaishnavim1121/MyMavenAnsibleWebApp1

Please enter Git repository.

Credentials

github credentials

Add

Advanced

Save Apply

Now we need to add a jenkinsfile into our project which will consist of the operations that will occur when building the project in Jenkins

It includes step wise instructions guiding at each stage with right commands

```

pipeline {
  agent any // Use any available agent

  environment {
    LANG = 'en_US.UTF-8'
    LC_ALL = 'en_US.UTF-8'
  } // this has to be added only if you get an error saying UTF required is 8 but showing in ISO8859

  tools {
    maven 'Maven' // Ensure this matches the name configured in Jenkins
  }

  stages {
    stage('Checkout') {
      steps {
        git branch: 'master', url: 'https://github.com/vaishnavimilli/MavenAnsibleWebApp1.git'
      }
    }

    stage('Build') {
      steps {
        sh 'mvn clean package' // Run Maven build
      }
    }

    stage('Archive') {
      steps {
        archiveArtifacts artifacts: 'target/*.war', fingerprint: true
      }
    }

    stage('Deploy') {
      steps {
        sh 'mvn clean package'
        sh 'ansible-playbook ansible/playbook.yml -i ansible/hosts.ini'
      }
    }
  }
}

```

Now try building the project with Jenkins

Error faced during build process:

The error happened because Ansible successfully copied the WAR file, but failed during the Tomcat restart step.

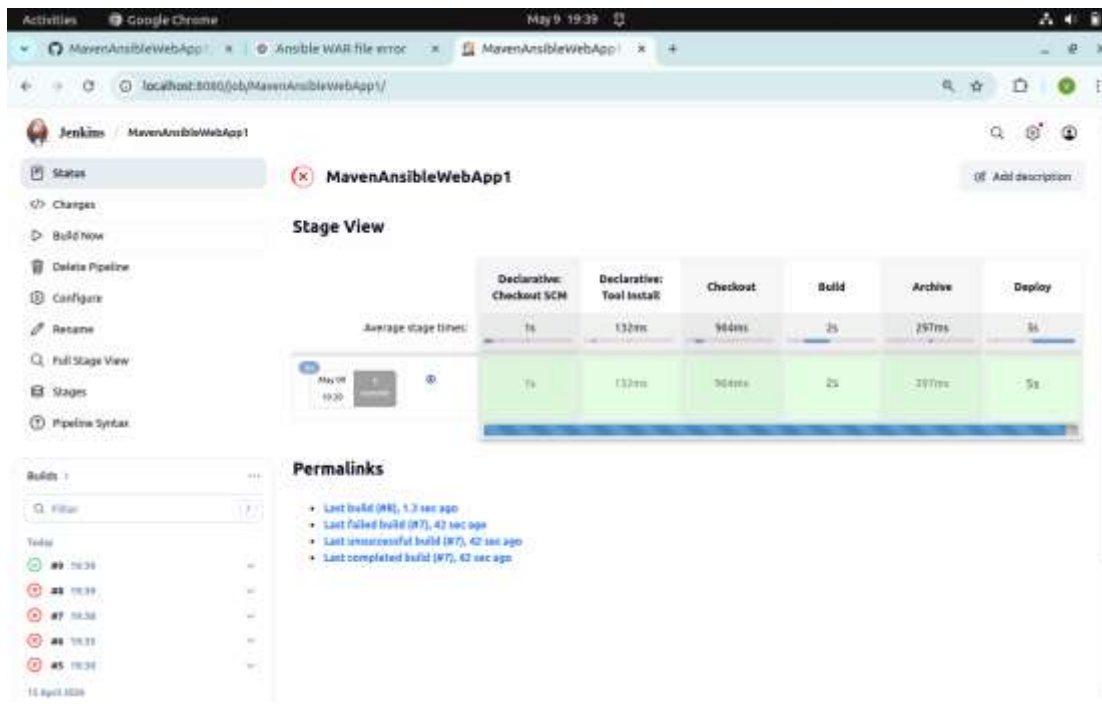
This means Jenkins/Ansible could not find a Linux service named tomcat using systemd.

We fixed it by removing the restart task from ansible/playbook.yml, since Tomcat automatically deploys WAR files placed inside:

/opt/tomcat/webapps/

After removing the restart section, the pipeline completed successfully.

Now try building the project again.



The screenshot displays the Jenkins web interface for a pipeline named 'MavenAnsibleWebApp1'. The pipeline is in a 'Completed' state, indicated by a green checkmark. The 'Stage View' shows a table of stages with their respective durations. The 'Permalinks' section lists recent builds, including a successful build (97) and several failed builds (99, 98, 96, 95).

Stage	Declarative: Checkout SCM	Declarative: Tool Install	Checkout	Build	Archive	Deploy
Average stage times:	1s	132ms	964ms	2s	197ms	3s
May 09 19:30	1s	132ms	964ms	2s	197ms	3s

Permalinks

- Last build (98), 1.3 sec ago
- Last failed build (97), 42 sec ago
- Last unsuccessful build (97), 42 sec ago
- Last completed build (97), 42 sec ago

The project is built completely.